



Tecnológico de Monterrey
Partial Exam 2 A
2026-Jan-May

Grade

Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 24/03/2026

Last Names and Names: _____

Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Multiple option section. Read the following questions and identify the option that best answers each, then write its letter on the line.

1. (4 points) _____ Identify the quotient rule for derivatives.

(a) $\left(\frac{u}{v}\right)' = \frac{v'u - u'v}{u^2}$ (b) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ (c) $\left(\frac{u}{v}\right)' = \frac{uv' - u'v}{v^2}$ (d) $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{u^2}$.

2. (4 points) _____ Identify the product rule for derivatives

(a) $(uv)' = u' + v'$ (b) $(uv)' = u'v + uv'$ (c) $(uv)' = uv' + uv'$ (d) $(uv)' = u'v + u'v$

3. (4 points) _____ Identify the chain rule for derivatives

(a) $[u(v(x))]' = v'(u(x))$ (b) $[u(v(x))]' = v'(u(x))u'(x)$
(c) $[u(v(x))]' = u'(v(x))v'(x)$ (d) $[u(v(x))]' = u'(v(x))$

4. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx}n^x = n^x \ln(n)$ (b) $\frac{d}{dx}n^x = \frac{1}{x}n^x$ (c) $\frac{d}{dx}n^x = xn^x$ (d) $\frac{d}{dx}n^x = ne^{x-1}$

5. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \sin(x) = \cos(x)$ (b) $\frac{d}{dx} \sin(x) = -\sin(x)$ (c) $\frac{d}{dx} \sin(x) = \sin(x)$ (d) $\frac{d}{dx} \sin(x) = -\cos(x)$

6. (4 points) _____ Which of the following derivatives is correct

(a) $\frac{d}{dx} \cos(x) = \cos(x)$ (b) $\frac{d}{dx} \cos(x) = -\sin(x)$ (c) $\frac{d}{dx} \cos(x) = \sin(x)$ (d) $\frac{d}{dx} \cos(x) = -\cos(x)$

7. (4 points) _____ Which of the following derivatives is correct

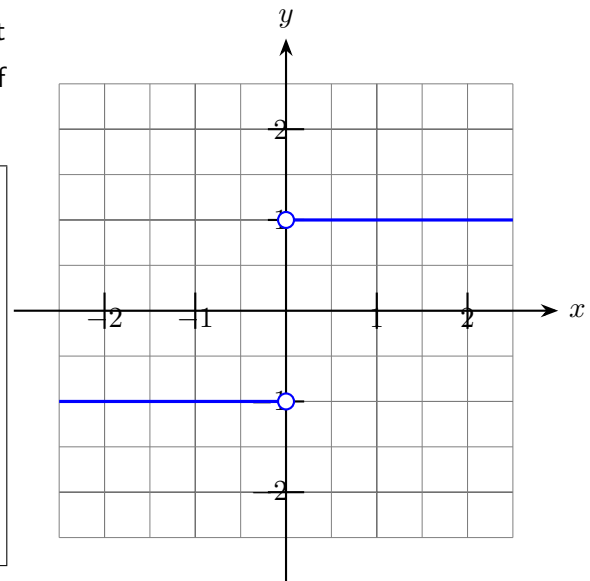
(a) $\frac{d}{dx} \log_n(x) = \frac{1}{\ln(x)}$ (b) $\frac{d}{dx} \log_n(x) = \frac{1}{x \ln(n)}$ (c) $\frac{d}{dx} \log_n(x) = \frac{\ln(x)}{x}$ (d) $\frac{d}{dx} \log_n(x) = \frac{1}{x \ln(x)}$

8. (4 points) _____ Which of the following options is the implicit derivative of $y^2 = x$

(a) $\frac{dx}{dy} = 0$ (b) $\frac{dx}{dy} = 2y/1$ (c) $\frac{dx}{dy} = 1$ (d) $\frac{dx}{dy} = 1/2y$

II Limits. Compute the following limits using the numeric, graphical or algebraic method.

- 1. (5 points)** Applying the graphical technique to compute the limit when $x \rightarrow 0$. Argument if the limit converges. Use the notion of limit from the left and right.



- 2. (5 points)** Compute the limit of the function $f(x) = \frac{\sin(x)}{x}$ as $x \rightarrow 0$ using the numeric method.

- 3. (5 points)** Compute the limit of the function $f(x) = \frac{6x^2+4}{5x^2-2}$ as $x \rightarrow \infty$ using the algebraic method.

4. (5 points) Compute the derivative of the function $f(x) = |x|$ at $x = 0$ using the definition of derivatives, $\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

III Derivatives by rules. Use the rules of derivatives that can be usefull to compute the derivatives of the following functions.

1. (8 points) Compute the derivative of the function $f(x) = e^{-x^2} \sin^2(x)$.

2. (8 points) Compute the derivative of the function $g(x) = \log [xe^{-x}]$.

IV Critical points. In this section your task is to find all critical points of the functions in the given context.

1. (8 points) An open box is made from a 5×5 cm sheet by cutting squares of side x from the corner. The volume of the box is given by $V(x) = x(5 - 2x)^2$. Find the length of the squares that achieves the greater volume.

2. (8 points) Find the critical point of the following function $f(x) = 6x \cdot e^{(3/2)x}$.

V Implicit derivatives.

1. (8 points) Compute the implicit derivative of the function $x^3 + y^3 = 6xy$.

IV High order derivatives. Compute the second order derivative of the following functions and reduce them as much as possible.

1. (8 points) $f(x) = e^{x+3}$